9 Programming a controller

This section explains the programming steps to setup a panel for a controls application. The steps are presented here in a somewhat logical order but the system is flexible, you can complete the sections you have information on as the data comes in and finish off the other parts later. It is assumed that at this point, the T3000 has been setup as per instructions in Chapters 3 and 8.

1. The Input and Output List

List all the inputs and outputs for the system to be controlled. A spreadsheet is generally available from the original project description. If you are starting from scratch, list the equipment and for each of these work out the list of inputs and outputs. Be sure to check the sequence of operations for the system, there's often wording in the text which doesn't match up well with the drawings, those few words can be expensive words which require extra sensors and i/o for your project.

Assuming you are doing your own list, its OK to proceed as information comes in, you can always revise the list as the project proceeds. The points don't have to be in any particular sequential order but its definitely best to keep related i/o near others on the same equipment. And certainly its best to keep i/o from one air handler for example, on one controller so that if the network goes down the equipment can still operate in stand alone mode. Software can connect any i/o to other i/o in the system but its best to keep the list in logical order.

-Once the list is together its time to assign some names to the i/o for entering into the system. In the T3000 Building Automation System we can we can assign each input or output with two names, a long one which is 20 characters for showing in reports and user screens, and a shorter one which is 8 characters to be used in the programming.

- Each input and output needs to be assigned to a physical input or output point on the controller. For inputs this is easy, any input can be jumper configured to match up with most any sensor, switch, transducer or meter. For the outputs, you need to assign the analog items to one of the analog outputs. On the binary outputs to enable fans and pumps the job is a little easier, assign each of these to an output relay on the controller.

TIP: If you need more binary outputs, a spare analog output can be converted to a binary output with an external 12VC coil relay, the analog output will drive the relay to gain one extra really from an analog output. The reverse is not true though, relays are fixed as binary outputs.

- The input list should show the sensor or transducer type and jumper settings.

- For valves and dampers especially, there are many arrangements for the i/o depending on the equipment and project specs so its best to have a data sheet for the exact item. The valves will be specified as normally open or closed, gather that information from the mechanical drawings and sequence description. The output signal can be floating control, modulating or on/off, some have feedback signal as well. All this info is in the data sheets for the actuator and is important to have on hand when laying out the system.

2. Configure “System” Components

Run the T3000 software and Sign-On with your Name and Password. Setup the T3000 as per instructions in Chapters 3 and 8.

3. Program Inputs

“Inputs” are the sensors and devices which are connected into the T3000 panel so that the T3000 can be programmed to act upon. For more information on In puts see section 6.3.

Select “Inputs” from the CONTROL menu, press ALT-I, or type INS from the command line.

Enter in a description that will identify each point. Each point must be unique or the system will not accept the description. When entering the 20 character “Description” as it is known, try to be as descriptive as possible. You should also develop a consistent nomenclature for the various types of control points such as “RTS” in front of all room temperature sensors. Using this nomenclature, the room sensor in office 102 would be named “RTS OFFICE102".

Also enter the “Label” which is the eight character short hand name for the input point which will be used in the Control Basic programs. Again, each LABEL must be unique or the system will not accept the name. By using the above nomenclature, the label for the same office temperature sensor would be “RTS102 ”.

Other Examples:

Input Point		Description	Label
Air Handling Unit 1	Supply Air Temperature	AHU1 SUPPLY AIR TEMP	AHU1.SAT
Air Handling Unit 1	Supply Fan Status	AHU1 SUPPLY FAN STAT	AHU1.S
Pump # 1 Radiation Loop Start/Stop	P1 RAD LOOP S/S	P1.S/S	

Select the range:

To set the range high light the field, press ENTER, and enter the desired range number. If no standard range will work with the sensor, see the section of this manual describing the construction of a “TABLE”.

The “D” field is a simple check mark field to verify the correct operation of each sensor. Toggle the field back to normal, check for correct readings, and enter calibration values if required.

Your changes are sent to the panel automatically if you leave the current screen, but to see the effect of current changes, you must Send and then Load the Input changes.

Input	Panel	Full Label	Auto/Manual	Value	Units	Range	Calibration	Sign	Filter	Status	Jumper	Label
IN1	1	SUPPLY AIR TEMP	Auto	-40.00	Deg.F	10K -40 to 250	0.0	-	5	Open	Thermistor Dry...	AHU1SAT
IN2	1	RETURN AIR TEMI	Auto	248.00	Deg.F	10K -40 to 250	0.0	-	5	Shorted	Thermistor Dry...	AHU1RAT
IN3	1	MIXED AIR TEMP	Auto	-40.00	Deg.F	10K -40 to 250	0.0	-	5	Open	Thermistor Dry...	AHUMAT
IN4	1	ROOM TEMP	Auto	-40.00	Deg.F	10K -40 to 250	0.0	-	5	Open	Thermistor Dry...	AHU1RT
IN5	1	FAN STATUS	Auto	On		Off/On			5	Normal	Thermistor Dry...	AHU1FS
IN6	1	OUTSIDE AIR TEMI	Auto	-40.00	Deg.F	10K -40 to 250	0.0	-	5	Open	Thermistor Dry...	OAT
IN7	1	IN 7	Auto	2.99		Unused	0.0	-	5	Normal	Thermistor Dry...	IN7
IN8	1	IN 8	Auto	2.99		Unused	0.0	-	5	Normal	Thermistor Dry...	IN8
IN9	1	IN 9	Auto	2.99		Unused	0.0	-	5	Normal	Thermistor Dry...	IN9
IN10	1	IN 10	Auto	2.99		Unused	0.0	-	5	Normal	Thermistor Dry...	IN10
IN11	1	IN 11	Auto	2.99		Unused	0.0	-	5	Normal	Thermistor Dry...	IN11
IN12	1	IN 12	Auto	2.99		Unused	0.0	-	5	Normal	Thermistor Dry...	IN12
IN13	1	IN 13	Auto	2.99		Unused	0.0	-	5	Normal	Thermistor Dry...	IN13
IN14	1	IN 14	Auto	2.99		Unused	0.0	-	5	Normal	Thermistor Dry...	IN14
IN15	1	IN 15	Auto	2.99		Unused	0.0	-	5	Normal	Thermistor Dry...	IN15
IN16	1	IN 16	Auto	2.99		Unused	0.0	-	5	Normal	Thermistor Dry...	IN16
IN17	1	IN 17	Auto	2.99		Unused	0.0	-	5	Normal	Thermistor Dry...	IN17
IN18	1	IN 18	Auto	0.00		Unused	0.0	-	5	Normal	Thermistor Dry...	IN18
IN19	1	IN 19	Auto	2.99		Unused	0.0	-	5	Normal	Thermistor Dry...	IN19
IN20	1	IN 20	Auto	2.99		Unused	0.0	-	5	Normal	Thermistor Dry...	IN20
IN21	1	IN 21	Auto	3.00		Unused	0.0	-	5	Normal	Thermistor Dry...	IN21
IN22	1	IN 22	Auto	3.00		Unused	0.0	-	5	Normal	Thermistor Dry...	IN22
IN23	1	IN 23	Auto	3.00		Unused	0.0	-	5	Normal	Thermistor Dry...	IN23
IN24	1	IN 24	Auto	3.00		Unused	0.0	-	5	Normal	Thermistor Dry...	IN24
IN25	1	IN 25	Auto	3.00		Unused	0.0	-	5	Normal	Thermistor Dry...	IN25
IN26	1	IN 26	Auto	3.00		Unused	0.0	-	5	Normal	Thermistor Dry...	IN26
IN27	1	IN 27	Auto	3.00		Unused	0.0	-	5	Normal	Thermistor Dry...	IN27
IN28	1	IN 28	Auto	3.00		Unused	0.0	-	5	Normal	Thermistor Dry...	IN28

4. Program Outputs

“Outputs” are the output signals from the T3000 sent to the fans, pumps, boilers, etc. which are connected to the T3000. Outputs can be either analog or digital. The software setting determines which mode the output will work with. The engineering units for each output can be configured to match the actual units normally associated with the device being controlled (i.e. Stop/Start , Close/Open, or % Open, % Closed, CFM). For more information on “Out-puts” see section 5.2.

To get to the outputs screen, type ALT-O or Select “Outputs” from the CONTROL menu. Alternatively, you can type OUTS from the command line.

Just as was described for the inputs, each OUT PUT re quires a 20 character description and an 8 character label. The names should follow a naming nomenclature similar to that used for the inputs, and each name must be unique.

To set the range highlight the field, press ENTER, and enter the desired range.

Enter in the Low and High volt ages for any analog devices such as valves or dampers which operate over 2-10 VDC range rather than the normal 0-10V range.

Enter starts delays as required. This is used for digital out puts only and de lays the start of a device in minutes: seconds up to a maximum delay of 99 minutes 50 seconds. A value entered in this field superimposes a lag on all other control actions such as weekly routines or basic programs which govern this point.

If you want to increase the security level on the output, enter that level in the “S” field.

Output	Panel	Full Label	Auto/Manual	HOA Switch	Value	Units	Range	PWM Period	Status	Label
OUT1	1	AHU1 FAN	Auto	AUTO	Off		Off/On	0	OK	AHU1FAN
OUT2	1	AHU1 COOLING	Auto	AUTO	Off		Off/On	0	-	AHU1COOL
OUT3	1	AHU1 HEATING	Auto	AUTO	Off		Off/On	0	OK	AHU1HEAT
OUT4	1	AHU1 MIX ↳?	Auto	AUTO	Off		Off/On	1	OK	
OUT5	1	┼	Manual	AUTO	0.00		Unused	32	OK	
OUT6	1		Auto	AUTO	0.00		Unused	0	OK	OUT6
OUT7	1	OUT7	Auto	AUTO	0.00		Unused	0	OK	OUT7
OUT8	1	OUT8	Auto	AUTO	0.00		Unused	0	OK	OUT8
OUT9	1	OUT9	Auto	AUTO	0.00		Unused	28	OK	OUT9
OUT10	1	E	Auto	AUTO	0.00		Unused	65	OK	?
OUT11	1		Auto	AUTO	0.00		Unused	0	OK	ACACAAA
OUT12	1	OUT12	Auto	AUTO	0.00		Unused	0	OK	OUT12
OUT13	1	OUT13	Auto	AUTO	0.00		Unused	0	OK	OUT13
OUT14	1	OUT14	Auto	AUTO	0.00		Unused	0	OK	OUT14
OUT15	1	OUT15	Manual	AUTO	0.00		Unused	168	OK	OUT1 ↳?
OUT16	1		Manual	AUTO	0.00		Unused	65	OK	
OUT17	1	CACACACACACAAA	Auto	AUTO	0.00		Unused	0	OK	OUT17
OUT18	1	OUT18	Auto	AUTO	0.00		Unused	0	OK	OUT18
OUT19	1	OUT19	Auto	AUTO	0.00		Unused	0	OK	OUT19
OUT20	1	OUT20	Auto	AUTO	0.00		Unused	0	OK	OUT20
OUT21	1	OUT21	Manual	AUTO	0.00		Unused	11	OK	g@
OUT22	1	?	Auto	AUTO	0.00		Unused	0	OK	
OUT23	1	OUT23	Auto	AUTO	0.00		Unused	0	OK	OUT23
OUT24	1	OUT24	Auto	AUTO	0.00		Unused	0	OK	OUT24
OUT25	1	OUT25	Auto	AUTO	0.00		Unused	0	-	OUT25
OUT26	1	OUT26	Auto	AUTO	0.00		Unused	0	-	OUT26
OUT27	1	6€	Manual	AUTO	688.38	Unused	Out of range	25	-	
OUT28	1	.e••	Auto	AUTO	0.00		Unused	0	-	OUT28

Toggle all outputs to be decommissioned. Later, as you verify their correct operation toggle the field back to normal.

When satisfied with their operation, Send and then Load the Outputs.

5. Define programming jobs based on out puts that you have to control

For example, a panel may have the following control jobs.

- Boiler Control
- Chiller Control
- Fan System 1 Control

Select “Programs” from the Control menu, or by typing PRGS from “Direct Access”. Enter in the program description and label. Each point must be unique or the system will not accept the description. When entering the description try to be as descriptive as possible.

For example: Air Handling Unit 1 Control = AHU1 PROGRAM

Program	Full Label	Status	Auto/Manual	Size	Run Status	Label
1	AHU1 PROGRAM	OFF	Auto	406	Normal	
2		OFF	Auto	64	Normal	
3		ON	Auto	29	Normal	
4		ON	Auto	0	Normal	
5		ON	Auto	0	Normal	
6		ON	Auto	0	Normal	
7		ON	Auto	0	Normal	
8		ON	Auto	0	Normal	
9		ON	Auto	0	Normal	
10		ON	Auto	0	Normal	
11		ON	Auto	0	Normal	
12		ON	Auto	0	Normal	
13		ON	Auto	0	Normal	
14		ON	Auto	0	Normal	
15		ON	Auto	0	Normal	
16		ON	Auto	0	Normal	

6. Write Individual Programs

Press the “Ins” key while highlighting the line of the de sired program, or using “Direct Access” type the program's label. As you write a program you will want to make use of 4 other point types:

- Variables - VARS
- Controllers - CONS
- Weekly Routines - WRS
- Annual Routines - ARS

Set up these points as the need for them arises. When finished writing your program, send the program by pressing “F2". This will first check the program for errors. If correct, it will send the program to the panel. If not, it will supply an error message complete with line numbers. For more information on Control Basic Programming see chapter 10.

7. Define Graphic Screens:

Graphic Screens are used to group points of all types into one logical unit for displaying on the screen. The elements of a group can be displayed over a graphic picture to augment the display of information.

Select “Graphic” from the Control menu. Enter in the name and label. Each label must be unique or the system will not accept the description. When entering the label try to be as descriptive as possible.

Graphic	Full Label	Label	Picture File	Refresh Rate
1	CACACACACABM	AHU1	radke	15
2				15
3				15
4				15
5		slO?		1
6			CACACAAA	0
7				15
8				15
9				15
10				101
11	◀¶ ├?			15
12				15
13				15
14				15
15				15
16		"u	?	1

8. Design screens:

Press the “Ins” key while highlighting the line of the desired screen.

To display a point on the screen, move the cursor to the desired location and press the “Ins” key. A pop-up box will appear on the screen to let you specify the element to be added to the display. A second pop- up menu will appear which controls the colour and size of the element text, and several other functions associated with this element. Press the “Esc” key to exit pop- up menu. Continue to add points to display until completion. When satisfied with the display go to File and Save.

For a more detailed description on building Screens see section 6.1.

9. Define Monitors

There are two types of monitors: Ana log which trends analog points, and Digital which logs start and stop times of digital points. Both are displayed together on a single monitor from the Data pull-down menu.

Set up monitors by entering the monitor name and entering the points to be monitored. Enter the interval time between samples, and then Send. Sampling of these points will now take place. If for some reason you decide to change some aspect of a monitor, it will erase all previous data and start a new one once you send the changes to the panel. For more information on Monitors see section 7.1.

10. Save the Program

Using the “File” selection on the Program menu, “Save” the panel program. Steps 1 through 9 are sent to the panel each time that you do a “Send”. By “Saving” the program, the program is saved in your PC and is available to be down loaded back into the panel if the panel crashes or loses its program.